



# Front-end Development

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## How to make things shiny?



# Contents

01

**Text-related Properties**

02

**Border**

03

**Background**

04

**Opacity**





# TEXT-SHADOW

`text-shadow: 3px 4px 5px #000;`

## 1. Text-related Properties

# Text-related CSS Properties

- **color** – specifies the color of the text. E.g., #fff (white), #000 (black), #ccc (border color), rgb(0, 0, 0), rgba(0, 0, 0, 0.5), blue, etc.
- **font-size** – size of font: xx-small, x-small, small, medium, large, x-large, xx-large, smaller, larger or **numeric value** (16px (desktop), 14px (mobile))
- **font-family** – comma separated font names
  - Example: verdana, sans-serif,...
  - The browser loads the **first one** that is **available**
  - There should always be **at least one generic font**:  
“serif”, “**sans-serif**”, “**CURSIVE**”, “*fantasy*”, “monospace”
- **font-weight** can be **normal** (400), **bold** (700), bolder, lighter (300) or a number in range [100 ... 900]

# Text-related CSS Properties

CSS Units: em, rem, and px

## ▪ Definition

- **px** (pixels): An absolute unit — fixed size that does not depend on parent elements.
- **em**: A relative unit based on the **font size** of the **parent** element. 1em = font-size of the parent.
- **rem**: A relative unit based on the font size of the root element (<html>). 1rem = font-size defined in the **html** tag.

# Text-related CSS Properties

CSS Units: em, rem, and px

- How em Works

```
body {  
    font-size: 16px;  
}  
p {  
    font-size: 1.5em;  
    /* 1.5 × 16px = 24px */  
}
```

```
/* If nested */  
div {  
    font-size: 20px;  
}  
div p {  
    font-size: 1.2em;  
    /* 1.2 × 20px = 24px */  
}  
div p span {  
    font-size: 1.2em;  
    /* 1.2 × 24px = 28.8px */  
}
```

The size **multiplies through parent levels**, so deep nesting can cause scaling issues.

# Text-related CSS Properties

CSS Units: em, rem, and px

- How rem Works

```
html {  
    font-size: 16px;  
}  
h1 {  
    font-size: 2rem;  
    /* 2 x 16px = 32px */  
}  
p {  
    font-size: 1rem;  
    /* 1 x 16px = 16px */  
}
```

Always consistent — does not depend on the parent element.

# Text-related CSS Properties

CSS Units: em, rem, and px

- Typical Font Scale Using em

```
body {  
    font-size: 1em;    /* Base = 16px */  
}  
h1 { font-size: 2.0em; } /* 32px */  
h2 { font-size: 1.5em; } /* 24px */  
h3 { font-size: 1.25em; } /* 20px */  
p { font-size: 1em; } /* 16px */  
small { font-size: 0.875em; } /* 14px */
```

# CSS Rules for Fonts

- **font-style** – styles the font  
Values: normal, italic
- **text-decoration** – decorates the text  
Values: none, underline, line-through, overline
- **text-align** – defines the alignment of text or other content  
Values: left, right, center, justify

# Shorthand Font Property

- **font**

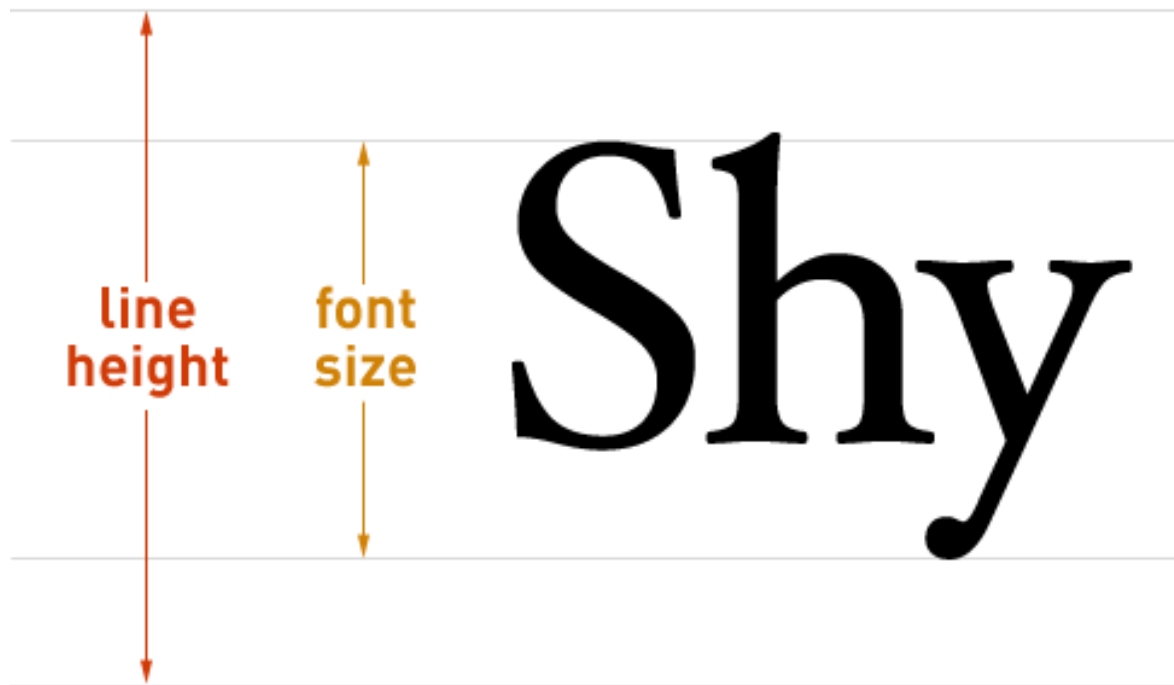
- Shorthand rule for setting multiple font properties at the same time

```
font: italic normal bold 12px/16px verdana;
```

- is equal to writing this:

```
font-style: italic;  
font-variant: normal;  
font-weight: bold;  
font-size: 12px;  
line-height: 16px;  
font-family: verdana;
```

# Shorthand Font Property



# Shorthand Font Property

## Line-height

- The line-height sets the **height of a line box**
- It's commonly used to set the distance between lines of text
- Values:
  - Normal
  - Number: 1, 1.5, 2,...
  - Length: 2em, 3em,...
  - Percentage: 34%, 200%,...

# Shorthand Font Property

Font-variant

- Normal

**Play with the two  
different font variants!**

- Small-caps

**PLAY WITH THE TWO  
DIFFERENT FONT VARIANTS!**



More Fonts

# Font Embeds

- Use font embedding instead of images
- Use **@font-face** to **declare** font
  - **src**: point to a font file on server
  - **font-family**: call the font



```
@font-face {  
    font-family: SketchRockwell;  
    src: url('SketchRockwell-Bold.ttf');  
}  
  
.my_CSS3_class {  
    font-family: SketchRockwell;  
    font-size: 3.2em;  
}
```

# Google Fonts



- **Link:** <https://fonts.google.com/>

- **Embed Font:**

- **<link>**

```
<link rel="preconnect" href="https://fonts.googleapis.com">  
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>  
<link href="https://fonts.googleapis.com/  
css2?family=Roboto:wght@400;700&display=swap"  
rel="stylesheet">
```

- **@import**

```
<style>  
  @import url('https://fonts.googleapis.com/  
    css2?family=Roboto:wght@400;700&display=swap');  
</style>
```

# Google Fonts

- Specify in CSS:

```
font-family: 'Roboto', sans-serif;
```

- **Note:**

Select the Language Font (Vietnamese)



# Text Shadow

- Applies shadow to text

- **Syntax:**

**text-shadow:** <horizontal-distance> <vertical-distance> <blur-radius> <shadow-color>;

- Do not alter the size of a box

- **Example:**

**text-shadow:** 2px 2px 7px #000000;

**This text is the preview text.**

# Text Overflow

- Specifies what should happen when text overflows the containing element
- **Syntax:** `text-overflow: <value>;`
- Possible **values**:
  - **ellipsis** - Display ellipses (...) to represent clipped text
  - **clip** - Default value, clips text
  - **string** - Display the given string to represent the clipped text (only works in Firefox)

This is some long text that ...

This is some long text that wi

# Text Overflow

- Both of the following **properties are required** for text-overflow:

- **white-space**: nowrap;
- **overflow**: hidden;

- Example:

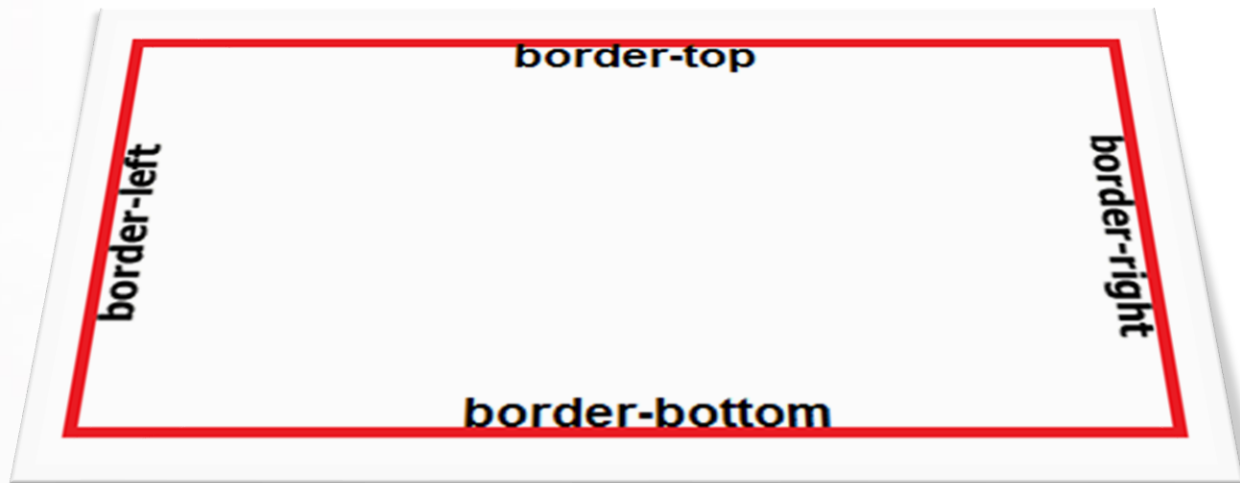
```
.text-overflow {  
    white-space: nowrap;  
    overflow: hidden;  
    text-overflow: ellipsis;  
    width: 100px;  
    height: 18px;  
    border: 1px solid #ddd;  
}
```

# Word Wrapping

- Allows long words to be able to be broken and wrap onto the next line
- **Syntax: word-wrap:** <value>;
- Possible **values**:
  - normal
  - break-word
- Supported in all major browsers

This paragraph has long words  
thisisaveryverylongwordthatisntre  
allyoneword and again a  
longwordwithnospacesinit

This paragraph has long words  
thisisaveryverylongwordthatisntreallyoneword  
and again a  
longwordwithnospacesinit



### 3. Borders

# Borders

- **border-width:** thin, medium, thick or numerical value (e.g. 10px)
- **border-color:** color alias (red, green,...) or RGB value
- **border-style:** none, hidden, dotted, dashed, solid, double, groove, ridge, inset, outset
- **Each property can be defined separately** for left, top, bottom and right  
border-top-style, border-left-color,...

# Border Shorthand Property

- **border**: shorthand rule for setting border properties at once:

```
border: 1px solid red;
```

- is equal to writing:

```
border-width: 1px;  
border-color: red;  
border-style: solid;
```

- Specify different borders for the sides via shorthand rules: **border-top**, **border-left**, **border-right**, **border-bottom**

# Box shadow

- Allows to easily implement multiple drop shadows (outer or inner) on box elements
- Specifying values for color, size, blur and offset
- **Syntax:**

**box-shadow:** [inset] offset-x offset-y [blur-radius] [spread-radius] color

- Example:

```
-moz-box-shadow: 10px 10px 5px #888;  
-webkit-box-shadow: 10px 10px 5px #888;  
box-shadow: 10px 10px 5px #888;
```

You can also define the color of the shadow. Here the shadow color is grey.

# Box shadow



The pulpit rock in Lysefjorden, Norway.



Monterosso al Mare. One of the five villages in Cinque Terre, Italy.

# Rounded Corners

- Rounded corners are a part of CSS 3
  - Supported in all major browsers
  - Firefox, IE 9, Chrome, Opera and Safari
- Done by the **border-radius** property

`border-radius: [<length>|<%>][<length>|<%>]?`

- Three ways to define corner radius:

`border-radius: 15px;`

`border-radius: 15px 20px;`

`border-radius: 15px 15px 15px 10px;`

# Rounded Corners

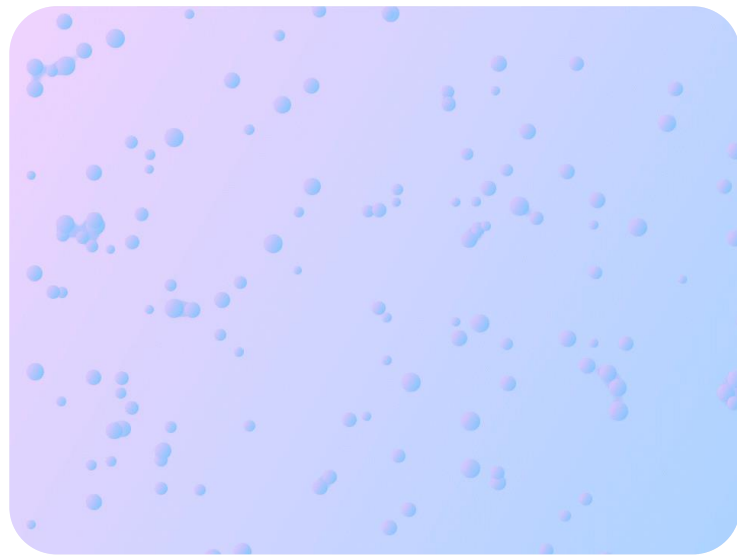
## Example

### **border-radius: 25px:**

The border-radius property defines the radius of the element's corners.

### **border-radius: 50px 20px:**

If two values are set; the first one is for the top-left and bottom-right corner, the second one for the top-right and bottom-left corner.



## 4. Background Properties

# Backgrounds

- **background-color**

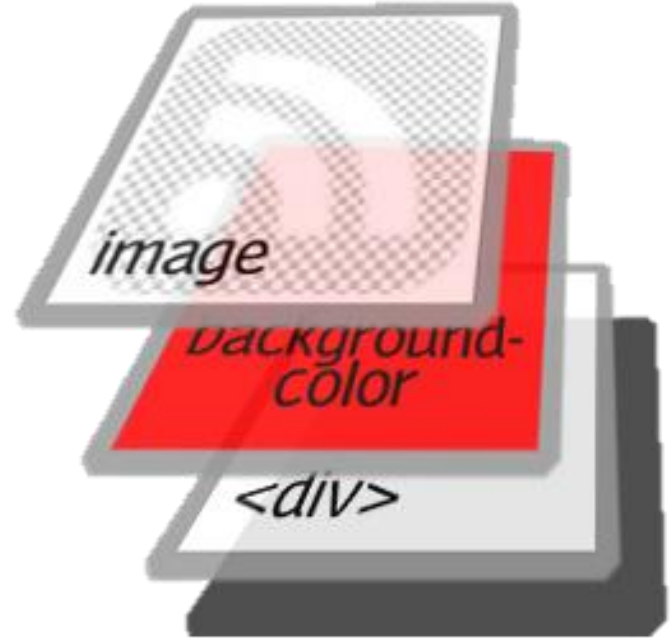
Set the background color of an element  
Can use color and image at the same time

- **background-image**

- URL of image to be used as background
- Example: `background-image: url("back.gif");`

- **background-repeat**

repeat-x, repeat-y, repeat, no-repeat



# Backgrounds

- **background-position**

- **specifies** vertical and horizontal **position of the background image**
- **Vertical** position: top, center, bottom
- **Horizontal** position: left, center, right
- Both can be specified in percentage or other numerical values
- Examples:

```
background-position: top left;  
background-position: -5px 50%;
```

# Backgrounds

## ▪ background-size

- **specifies** the **size** of the **background images**
- **Value:** auto | length | contain | cover

|         |                                                                                                                                         |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------|
| auto    | Default value. The background image is displayed in its original size                                                                   |
| length  | Sets the width and height of the background image.                                                                                      |
| cover   | Resize the background image to cover the entire container, even if it has to stretch the image or cut a little bit off one of the edges |
| contain | Resize the background image to make sure the image is fully visible                                                                     |

- Examples: `background: url(mountain.jpg);`  
`background-repeat: no-repeat;`  
`background-size: 75% 50%;`

# Background Shorthand Property

- **background**: shorthand rule for setting background properties at the same time:

```
background: #FFF0C0 url("back.gif") no-repeat fixed top;
```

- is equal to writing:

```
background-color: #FFF0C0;  
background-image: url("back.gif");  
background-repeat: no-repeat;  
background-attachment: fixed;  
background-position: top;
```

- Some browsers will not apply BOTH color and image for background if using shorthand rule

# Background-image or <img>?

- **Background images** allow you to **save many image tags** from the HTML
  - Leads to less code
  - More content-oriented approach
- All **images** that are not part of the page content (and are used only for “**beautification**”) should be **moved to the CSS**

# Gradient Backgrounds

- Gradients are **smooth transitions** between two or more specified colors
- Use of CSS gradients can replace images and reduce download time
  - Lots of gradient generators on the WEB
- Create a more flexible layout, and **look better** while **zooming**
- Supported in all major browsers via different keywords



# Gradient Backgrounds

## Example

```
background: rgb(34,193,195);  
background: -moz-linear-gradient(0deg,  
    rgba(34,193,195,1) 0%, rgba(253,187,45,1) 100%);  
background: -webkit-linear-gradient(0deg,  
    rgba(34,193,195,1) 0%, rgba(253,187,45,1) 100%);  
background: linear-gradient(0deg,  
    rgba(34,193,195,1) 0%, rgba(253,187,45,1) 100%);
```





## Contact Form

Nunc diam leo, fringilla vulputate elit lobortis, consectetur vestibulum quam. Sed id felis ligula. In euismod libero at magna dapibus, in rutrum velit lacinia. Etiam a mi quis arcu varius condimentum.

Name

Email

Subject

Your message here...

## 5. Opacity

# Opacity

- **opacity**

specifies the opacity of the element

- Floating point number from **0** to **1**
- For old Mozilla browsers use **–moz-opacity**
- For IE use **filter:alpha(opacity=value)** where value is from 0 to 100;

# Opacity

## Using the opacity property:

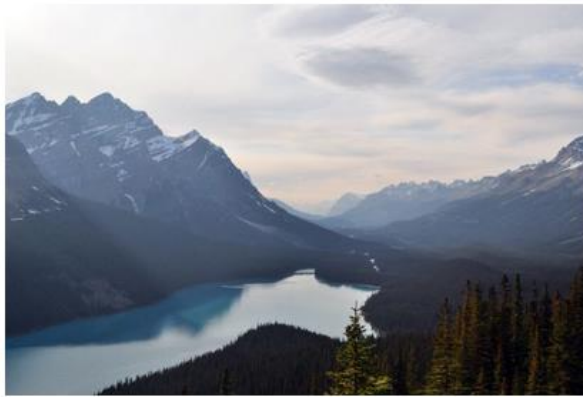


## Using RGBA color values:



# Opacity

The **opacity** property is often used together with the **:hover** selector to change the opacity on mouse-over:





# Summary

# Exercise

Write CSS for the Lani & Dani webpage.





The background image shows a clean, white desk with various items: a teal desk lamp on the left, a small green succulent, a stack of papers, a large black monitor on the right, a keyboard, a mouse, and a small teal box with a pen and a cup. A large, semi-transparent teal circle is centered over the desk, containing the text 'Q&A' in a bold, black, sans-serif font.

**Q&A**



Read More

# Remove List-Item Markers

The **list-style-type:none;** property is used to remove the list-item markers.

```
ul {  
    list-style-type: none;  
}
```

# Variables - var() Function

## Declare a Variable

- To create a global variable, declare it inside the **:root** selector. The :root selector matches the document's root element.
- A CSS variable name must begin with two dashes (--) and is case sensitive!
- E.g.:

```
:root {  
    --primary-bg-color: green;  
}
```

# Variables - var() Function

## var() Function

- The **var()** function is used to insert the value of a declared CSS variable.
- **Syntax:** var(--name[, value])
  - Name: Required. The variable name (must start with two dashes)
  - Value: Optional. The fallback value (used if the variable is not found)

- E.g.:

```
:root {  
    --primary-bg-color: #1e90ff;  
    --primary-color: #ffffff;  
}  
  
body {  
    background-color: var(--primary-bg-color);  
}
```